

## Branch: problemSet1

### 1. Boot xv6 -- setting up the repository and 'ls' command

```
arwam@Arwa: /mnt/c/Users/i/ x + v
ballocc: first 1128 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2489
findtest.sh 2 3 97
sixfive.txt 2 4 24
cat        2 5 36760
echo       2 6 35608
forktest   2 7 17384
grep       2 8 40192
init       2 9 36064
kill       2 10 35552
ln         2 11 35344
ls         2 12 38888
mkdir      2 13 35592
rm         2 14 35584
sh         2 15 58608
stressfs   2 16 36456
usertests  2 17 201568
grind      2 18 52016
wc         2 19 37744
zombie     2 20 34944
logstress  2 21 37528
forphan    2 22 36344
dorphan    2 23 35792
sync       2 24 35008
sleep      2 25 35016
sixfive    2 26 37544
find       2 27 40040
memdump    2 28 40816
console    3 29 0
```

### 2. Sleep (pauses for a few seconds, no output) and sixfive

```
arwam@Arwa: /mnt/c/Users/i/ x + v
hart 1 starting
hart 2 starting
init: starting sh
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2489
findtest.sh 2 3 97
sixfive.txt 2 4 24
cat        2 5 36760
echo       2 6 35608
forktest   2 7 17384
grep       2 8 40192
init       2 9 36064
kill       2 10 35552
ln         2 11 35344
ls         2 12 38888
mkdir      2 13 35592
rm         2 14 35584
sh         2 15 58608
stressfs   2 16 36456
usertests  2 17 201568
grind      2 18 52016
wc         2 19 37744
zombie     2 20 34944
logstress  2 21 37528
forphan    2 22 36344
dorphan    2 23 35792
sync       2 24 35008
sleep      2 25 35016
sixfive    2 26 37544
find       2 27 40040
memdump    2 28 40816
console    3 29 0
$ sleep 10
$ sixfive sixfive.txt
5
100
18
6
$ memdump
```

### 3. Memdump (all commands starting with memdump) and Find (starting with echo commands onwards)

```
arwam@Arwa: /mnt/c/Users/ / x + v
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
000000000000d18
1819438967
100
z
xyzzy
Example 5:
hello
w
o
r
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
exec echo failed
memdump: not enough data for 'p'
$ echo deadc0de | memdump p
6564306364616564
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
exec echo failed
$ echo > a/aa/b
$ find . b
```

```
arwam@Arwa: /mnt/c/Users/ / x + v
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
exec echo failed
memdump: not enough data for 'p'
$ echo deadc0de | memdump p
6564306364616564
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
exec echo failed
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ $ $ $ $ $ hello
hello
hello
$ $ $ echo done
done
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ mkdir: c failed to create
$ $ $ $ $ hello
hello
hello
$ $ $ QEMU: Terminated
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ clean qemu
Command 'clean' not found, did you mean:
```

#### 4. Exec – sh < findtest.sh

```
o
r
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
exec echo failed
memdump: not enough data for 'p'
$ echo deadc0de | memdump p
6564306364616564
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
exec eco failed
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ $ $ $ $ $ $ hello
hello
hello
$ $ $ echo done
done
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ $ $ $ $ $ $ hello
hello
hello
$ $ $ QEMU: Terminated
```

## GRADE TESTS FOR PSET1:

### 1. Sleep

```
arwam@Arwa: /mnt/c/Users/...
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Documents/Gith
ub/xv6-labs-2026-. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin
-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-built
in-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump | sed '1,/SYMBOL TABLE/d; s/ .* //; /*$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/
_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1166 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ uptime
265
$ QEMU: Terminated
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util uptime
make: 'kernel/kernel' is up to date.
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (4.2s)
== Test sleep, returns == sleep, returns: OK (2.0s)
== Test sleep, makes syscall == sleep, makes syscall: OK (2.1s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (2.0s)
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$
```

### 2. Sixfive

```
arwam@Arwa: /mnt/c/Users/  ×  +  v
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Documents/Gith
ub/xv6-labs-2026: -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin
-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-built
in-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/_memdump.o user/_ulib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/_memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /*$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/
_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1166 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ uptime
 265
$ QEMU: Terminated
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util uptime
make: 'kernel/kernel' is up to date.
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (4.2s)
== Test sleep, returns == sleep, returns: OK (2.0s)
== Test sleep, makes syscall == sleep, makes syscall: OK (2.1s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (2.0s)
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sixfive
make: 'kernel/kernel' is up to date.
== Test sixfive_test == sixfive_test: OK (3.5s)
== Test sixfive_readme == sixfive_readme: OK (2.8s)
== Test sixfive_all == sixfive_all: OK (2.9s)
== Test sixfive_stdin and missing file == sixfive, stdin and missing file: OK (2.3s)
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$
```

### 3. Memdump

```
arwam@Arwa: /mnt/c/Users/  ×  +  v
-I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/_memdump.o user/_ulib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/_memdump.asm
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/
_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1166 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ uptime
 265
$ QEMU: Terminated
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util uptime
make: 'kernel/kernel' is up to date.
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (4.2s)
== Test sleep, returns == sleep, returns: OK (2.0s)
== Test sleep, makes syscall == sleep, makes syscall: OK (2.1s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (2.0s)
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sixfive
make: 'kernel/kernel' is up to date.
== Test sixfive_test == sixfive_test: OK (3.5s)
== Test sixfive_readme == sixfive_readme: OK (2.8s)
== Test sixfive_all == sixfive_all: OK (2.9s)
== Test sixfive_stdin and missing file == sixfive, stdin and missing file: OK (2.3s)
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util memdump
make: 'kernel/kernel' is up to date.
== Test memdump, examples == memdump, examples: OK (4.0s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (2.3s)
== Test memdump, formats h c, bounds checking, usage == memdump, formats h c, bounds checking, usage: OK (2.4s)
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$
```

### 4. Find

```
arwam@Arwa: /mnt/c/Users/...  
_sixfive user/_find user/_uptime user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1166 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no  
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 2 starting  
hart 1 starting  
init: starting sh  
$ uptime  
265  
$ QEMU: Terminated  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util uptime  
make: 'kernel/kernel' is up to date.  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sleep  
make: 'kernel/kernel' is up to date.  
== Test sleep, no arguments == sleep, no arguments: OK (4.2s)  
== Test sleep, returns == sleep, returns: OK (2.0s)  
== Test sleep, makes syscall == sleep, makes syscall: OK (2.1s)  
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (2.0s)  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util sixfive  
make: 'kernel/kernel' is up to date.  
== Test sixfive.test == sixfive.test: OK (3.5s)  
== Test sixfive.readme == sixfive.readme: OK (2.8s)  
== Test sixfive.all == sixfive.all: OK (2.9s)  
== Test sixfive, stdin and missing file == sixfive, stdin and missing file: OK (2.3s)  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util memdump  
make: 'kernel/kernel' is up to date.  
== Test memdump, examples == memdump, examples: OK (4.0s)  
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (2.3s)  
== Test memdump, formats h c, bounds checking, usage == memdump, formats h c, bounds checking, usage: OK (2.4s)  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ ./grade-lab-util find  
make: 'kernel/kernel' is up to date.  
== Test find, in current directory == find, in current directory: OK (4.5s)  
== Test find, in sub-directory == find, in sub-directory: OK (2.7s)  
== Test find, recursive == find, recursive: OK (2.9s)  
== Test find, missing target, no duplicates, directory match == find, missing target, no duplicates, directory match: OK (3.1s)  
arwam@Arwa:/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ |
```

## Branch: Assignment1

### 1. Uptime

```
arwam@Arwa: /mnt/c/Users/...  
mv -f fs.img fs.img.bk  
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Do  
cuments/Github/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=riscv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-buil  
tin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-  
bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-p  
rintf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sh.o user/sh.c  
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_sh user/sh.o user/ulib.o user/usys.o user/printf.o user/umalloc.o  
riscv64-unknown-elf-objdump -S user/_sh > user/sh.asm  
riscv64-unknown-elf-objdump -t user/_sh | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sh.sym  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls  
user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user  
/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1166 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=  
fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 2 starting  
hart 1 starting  
init: starting sh  
$ uptime  
165  
$
```

### 2. regex

```
arwam@Arwa: /mnt/c/Users/...
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ nano user/find.c
arwam@Arwa: /mnt/c/Users/arwam/Documents/Github/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
make: [Makefile:293: newfs.img] Error 1 (ignored)
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/find.o user/find.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_find > user/find.asm
riscv64-unknown-elf-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1167 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . b
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$
```

3. modify the shell to not print a \$ when processing shell commands from a file ( it can be compared against the previous output of exec that many '\$' were printed, which is not the case anymore indicating our changed to have worked)

```
arwam@Arwa: /mnt/c/Users/...
riscv64-unknown-elf-objdump -S user/_find > user/find.asm
riscv64-unknown-elf-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/uptime.o user/uptime.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_uptime user/uptime.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_uptime > user/uptime.asm
riscv64-unknown-elf-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/Documents/Github/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1166 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sh < findtest.sh
hello
hello
hello
$
```